Introduction to Applied Cryptography – LDACS1310
Exercises Session 7

Questions

Question 1: ElGamal Public Key Encryption and CCA Security

1. Write the security definition of CCA security for a public key encryption scheme.
2. Let (c_1, c_2) and (c'_1, c'_2) be ElGamal encryptions, with the same public key, of plaintexts m and m' respectively. Is $(c_1 c'_1, c_2 c'_2)$ a valid ciphertext w.r.t. the same public key? If yes, what is its decryption?
3. Check the property observed in the previous question in the code file `elgamal.py` and complete the missing parts.
4. Given an encryption (c_1, c_2) of m, can you build part valid encryption of m, knowing the public key but not m? (Remember that the public key is $(\mathbb{G}, g, q, h = g^x)$)
5. Check the property observed in the previous question in the code file `elgamal.py` and complete the missing parts.
6. Show that ElGamal encryption is not CCA-secure.

Question 2: A Variation of ElGamal

Let us consider the ElGamal public encryption scheme modified to encrypt messages in $\mathcal{M} = \{0, 1\}$ with the encryption algorithm $\mathsf{Enc}_{(\mathbb{G},q,g,h)}(b)$: choose independent uniform $y, z \in \mathbb{Z}_q$, then set $c_1 = g^y$ and $c_2 = h^y$ if $b = 1$, while $c_2 = g^z$ if $b = 0$. Output the ciphertext (c_1, c_2).

1. How is it possible to decrypt correctly such ciphertexts, knowing the private key?
2. Show that this scheme is CPA-secure if DDH holds in $\mathbb{G}$.

Question 3: DDH PRG

Let $\mathbb{G}$ be a cyclic group of prime order q generated by $g \in \mathbb{G}$. Consider the following PRG defined over $(\mathbb{Z}_q^2, \mathbb{G}^3)$: $G(\alpha, \beta) := (g^\alpha, g^\beta, g^{\alpha\beta})$. Define what it means for a PRG over $(\mathbb{Z}_q^2, \mathbb{G}^3)$ to be secure and show that G is a secure PRG assuming that DDH holds in $\mathbb{G}$.